

Double integrals over rectangular area

Eg1-A. from Tim's note: Evaluate $\int x^2 + 4xy + y^2 - 12y + 2 dx$.

when integrating with respect to x , y is like a number.

$$\int x^2 + 4xy + y^2 - 12y + 2 dx = \frac{x^3}{3} + 2x^2y + y^2x - 12yx + 2x + C$$

Eg1-B. from Tim's note: Evaluate $\int x^2 + 4xy + y^2 - 12y + 2 dy$.

when integrating with respect to y , x is like a number.

$$\int x^2 + 4xy + y^2 - 12y + 2 dy = x^2y + 2xy^2 + \frac{y^3}{3} - 6y^2 + 2y + C$$

Eg2-A. from Tim's note: Evaluate $\int_0^1 \int_1^2 x^2 + 4xy + y^2 - 12y + 2x \, dx \, dy$.

$$\begin{aligned} & \int_0^1 \left. \frac{x^3}{3} + 2x^2y + y^2x - 12yx + 2x \right|_1^2 dy \\ &= \int_0^1 \left[\frac{8}{3} + 8y + 2y^2 - 24y + 4 - \frac{1}{3} - 2y - y^2 + 12y - 2 \right] dy \\ &= \int_0^1 \left(\frac{13}{3} - 6y + y^2 \right) dy = \left. \frac{13}{3}y - 3y^2 + \frac{y^3}{3} \right|_0^1 \\ &= \frac{13}{3} - 3 + \frac{1}{3} = \frac{14-9}{3} = \frac{5}{3} \end{aligned}$$

Eg2-B. from Tim's note: Evaluate $\int_1^2 \int_0^1 x^2 + 4xy + y^2 - 12y + 2 dy dx$.

$$\int_1^2 x^2 y + 2xy^2 + \frac{y^3}{3} - 6y^2 + 2y \Big|_0^1 dx$$

$$= \int_1^2 x^2 + 2x + \frac{1}{3} - 6 + 2 dx = \int_1^2 x^2 + 2x - \frac{11}{3} dx$$

$$= \frac{x^3}{3} + x^2 - \frac{11}{3}x \Big|_1^2 = \left(\frac{8}{3} + 4 - \frac{22}{3}\right) - \left(\frac{1}{3} + 1 - \frac{11}{3}\right)$$

$$= -\frac{2}{3} + \frac{7}{3} = \frac{5}{3}$$

Eg3. from Tim's note: Find the double integral $\int \int x \sqrt{2x^2 + 3y} dy dx$ the rectangular region $0 \leq x \leq 1, 1 \leq y \leq 2$.

$$\int_0^1 \int_1^2 x (2x^2 + 3y)^{1/2} dy dx$$

$$v = 2x^2 + 3y \Rightarrow \frac{dv}{dy} = 3 \Rightarrow dv = 3 dy$$

$$y=1 \Rightarrow v = 2x^2 + 3$$

$$y=2 \Rightarrow v = 2x^2 + 6$$

$$\frac{1}{3} \int_0^1 \int_{2x^2+3}^{2x^2+6} x v^{1/2} dv dx = \frac{1}{3} \int_0^1 x \cdot \frac{2}{3} \cdot v^{3/2} \Big|_{2x^2+3}^{2x^2+6} dx$$

$$= \frac{2}{9} \int_0^1 x \left[(2x^2+6)^{3/2} - (2x^2+3)^{3/2} \right] dx$$

$$\begin{aligned}
& \frac{2}{9} \int_0^1 x(2x^2+6)^{3/2} dx - \frac{2}{9} \int_0^1 x(2x^2+3)^{3/2} dx \\
&= \frac{2}{9} \cdot \frac{1}{4} \int_0^1 4x(2x^2+6)^{3/2} dx - \frac{2}{9} \cdot \frac{1}{4} \int_0^1 4x(2x^2+3)^{3/2} dx \\
&= \frac{1}{18} \frac{2}{5} (2x^2+6)^{5/2} \Big|_0^1 - \frac{1}{18} \frac{2}{5} (2x^2+3)^{5/2} \Big|_0^1 \\
&= \frac{1}{45} (2x^2+6)^{5/2} \Big|_0^1 - \frac{1}{45} (2x^2+3)^{5/2} \Big|_0^1 \\
&= \frac{1}{45} \left[8^{5/2} - 6^{5/2} - 5^{5/2} + 3^{5/2} \right]
\end{aligned}$$

Eg 4. from Tim's note: Find the volume under the surface $z = \frac{1}{xy}$ and above the rectangular region $1 \leq x \leq e$, $1 \leq y \leq e$.

$$\begin{aligned}\int_1^e \int_1^e \frac{1}{xy} dx dy &= \int_1^e \frac{dx}{x} \int_1^e \frac{dy}{y} \\ &= \ln|x| \Big|_1^e \cdot \ln|y| \Big|_1^e \\ &= (\ln e - \ln 1) \cdot (\ln e - \ln 1) \\ &= 1 \cdot 1 = 1\end{aligned}$$

Eg 5. from Tim's note: Find the volume under the surface $z = e^{2x-3y}$ and above the rectangular region $0 \leq x \leq 1$, $0 \leq y \leq 2$.

$$\int_0^2 \int_0^1 e^{2x-3y} dx dy$$

$$= \int_0^2 \int_0^1 e^{2x} \cdot e^{-3y} dx dy$$

$$= \int_0^1 e^{2x} dx \int_0^2 e^{-3y} dy = \left. \frac{e^{2x}}{2} \right|_0^1 \cdot \left. \frac{e^{-3y}}{-3} \right|_0^2$$

$$= -\frac{1}{6} (e^2 - 1) (e^{-6} - 1) = -\frac{1}{6} (e^{-4} - e^{-6} - e^2 + 1)$$